

DBMS: ASSIGNMENT 8

MongoDB- Aggregation and Indexing

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Case Study: Staff Metrics Analytics Store

```
test> use staffMetricsHub
switched to db staffMetricsHub
```

```
staffMetricsHub> db.staff.insertMany([
| { staff_id: 1, name: "Adi Maitre", salary: 10000, rating: 5 },
| { staff_id: 2, name: "Kritika Chaudhari", salary: 15000, rating: 4 },
| { staff_id: 3, name: "Ritobrata Baksi", salary: 20000, rating: 3 },
| { staff_id: 4, name: "Soham Kulkari", salary: 18000, rating: 2 }
| ])
{
  acknowledged: true
}
```

```
staffMetricsHub> db.staff.find()
```

```
staffMetricsHub> db.staff.aggregate([
| { $group: { _id: null, total_salary: { $sum: "$salary" } } }
| ])
[ { _id: null, total_salary: 63000 } ]
```

```
staffMetricsHub> db.staff.aggregate([
| { $group: { _id: null, avg_salary: { $avg: "$salary" } } }
| ])
[ { _id: null, avg_salary: 15750 } ]
```

```
staffMetricsHub> db.staff.aggregate([
| { $group: { _id: null, max_salary: { $max: "$salary" } } }
| ])
[ { _id: null, max_salary: 20000 } ]
```

```
staffMetricsHub> db.staff.aggregate([
| { $group: { _id: "$rating", total_salary: { $sum: "$salary" } } }
| ])
```

```
staffMetricsHub> db.staff.createIndex({ salary: 1 })
```

salary_1

```
staffMetricsHub> db.staff.createIndex({ rating: -1 })  
rating_-1
```

```
staffMetricsHub> db.staff.getIndexes()
```

Project: Smart Tax Analytics Document Store

test> use smartTaxAnalytics

switched to db smartTaxAnalytics

```
smartTaxAnalytics> db.user_login.insertMany([  
  | { user_id: 1, first_name: "Adi", last_name: "Maitre", username: "adi123", role: "taxpayer" },  
  | { user_id: 2, first_name: "Kritika", last_name: "Chaudhari", username: "kritika123", role:  
  "taxpayer" },  
  | { user_id: 3, first_name: "Ritobrata", last_name: "Baksi", username: "rito123", role: "taxpayer" },  
  | { user_id: 4, first_name: "Sarang", last_name: "Kolekar", username: "sarang123", role:  
  "taxpayer" }  
  | ])
```

```
smartTaxAnalytics> db.taxpayer_profile.insertMany([  
  | { taxpayer_id: 1, user_id: 1, pan: "PAN001", city: "Pune" },  
  | { taxpayer_id: 2, user_id: 2, pan: "PAN002", city: "Mumbai" },  
  | { taxpayer_id: 3, user_id: 3, pan: "PAN003", city: "Delhi" },  
  | { taxpayer_id: 4, user_id: 4, pan: "PAN004", city: "Nashik" }  
  | ])
```

```
smartTaxAnalytics> db.income_sources.insertMany([  
  | { taxpayer_id: 1, amount: 600000, year: "2024-25" },  
  | { taxpayer_id: 2, amount: 300000, year: "2024-25" },  
  | { taxpayer_id: 3, amount: 800000, year: "2024-25" },  
  | { taxpayer_id: 4, amount: 200000, year: "2024-25" }  
  | ])
```

```
smartTaxAnalytics> db.deductions.insertMany([  
  | { taxpayer_id: 1, amount: 50000 },  
  | { taxpayer_id: 2, amount: 20000 },  
  | { taxpayer_id: 3, amount: 60000 },  
  | { taxpayer_id: 4, amount: 10000 }  
  | ])
```

```
smartTaxAnalytics> db.income_sources.find()
```

```
smartTaxAnalytics> db.income_sources.aggregate([
| { $group: { _id: null, total_income: { $sum: "$amount" } } }
| ])
[ { _id: null, total_income: 1900000 } ]
```

```
smartTaxAnalytics> db.income_sources.aggregate([
| { $group: { _id: null, avg_income: { $avg: "$amount" } } }
| ])
[ { _id: null, avg_income: 475000 } ]
```

```
smartTaxAnalytics> db.income_sources.aggregate([
| { $group: { _id: "$taxpayer_id", total_income: { $sum: "$amount" } } },
| { $match: { total_income: { $gt: 500000 } } },
| { $project: { total_income: 1, tax: { $multiply: ["$total_income", 0.1] } } }
| ])
```

```
smartTaxAnalytics> db.taxpayer_profile.aggregate([
| {
|   $lookup: {
|     from: "income_sources",
|     localField: "taxpayer_id",
|     foreignField: "taxpayer_id",
|     as: "income_details"
|   }
| }
| ])
```

```
smartTaxAnalytics> db.user_login.createIndex({ username: 1 })
username_1
```

```
smartTaxAnalytics> db.income_sources.createIndex({ taxpayer_id: 1 })
taxpayer_id_1
```

```
smartTaxAnalytics> db.income_sources.createIndex({ taxpayer_id: 1, year: 1 })
taxpayer_id_1_year_1
```

```
smartTaxAnalytics> db.income_sources.getIndexes()
```